

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 15-2022: A 57-Year-Old Man with Persistent Cough and Pulmonary Opacities

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and Lida P. Hariri, M.D., Ph.D.

PRESENTATION OF CASE

Dr. George A. Alba: A 57-year-old man was evaluated at this hospital because of persistent cough and pulmonary opacities.

Six years before the current admission, the patient received a diagnosis of polyarteritis nodosa after fevers, arthralgias, abdominal pain, and testicular pain had developed. He was treated with six cycles of cyclophosphamide, followed by maintenance therapy with prednisone and weekly methotrexate. He was treated intermittently with escalating doses of prednisone because of episodic worsening of the arthralgias; the most recent such treatment was given 5 months before the current admission. At that time, owing to an exacerbation of pain associated with the arthralgias and at the direction of his physician, he increased the amount of oxycodone he had been taking. He also reported fever and weight loss. Imaging studies were obtained.

Dr. Jeanna M. Harvey Barnes: Five months before the current admission, computed tomography (CT) of the chest, abdomen, and pelvis was performed after the administration of intravenous contrast material. Chest CT (Fig. 1A and 1B) revealed mild centrilobular emphysema and small pulmonary nodules that appeared unchanged from those seen on imaging that had been performed 6 years earlier. There was also diffuse thickening of the bronchial wall, layering secretions in the trachea and right main-stem bronchus, a mildly patulous esophagus, and a small hiatal hernia. There were no clinically significant findings on imaging of the abdomen or pelvis.

Dr. Alba: A test for *Mycobacterium tuberculosis* was negative. Serum protein electrophoresis results were normal, as was the IgG level.

Three months before the current admission, the patient had fevers and a cough productive of green sputum. The temperature was 37.3°C and the oxygen saturation 89% while he was breathing ambient air. Thrush was present in the oropharynx. Auscultation of the chest revealed crackles and wheezing in the right lung. Blood levels of electrolytes, lipase, amylase, lactate dehydrogenase, and troponin T

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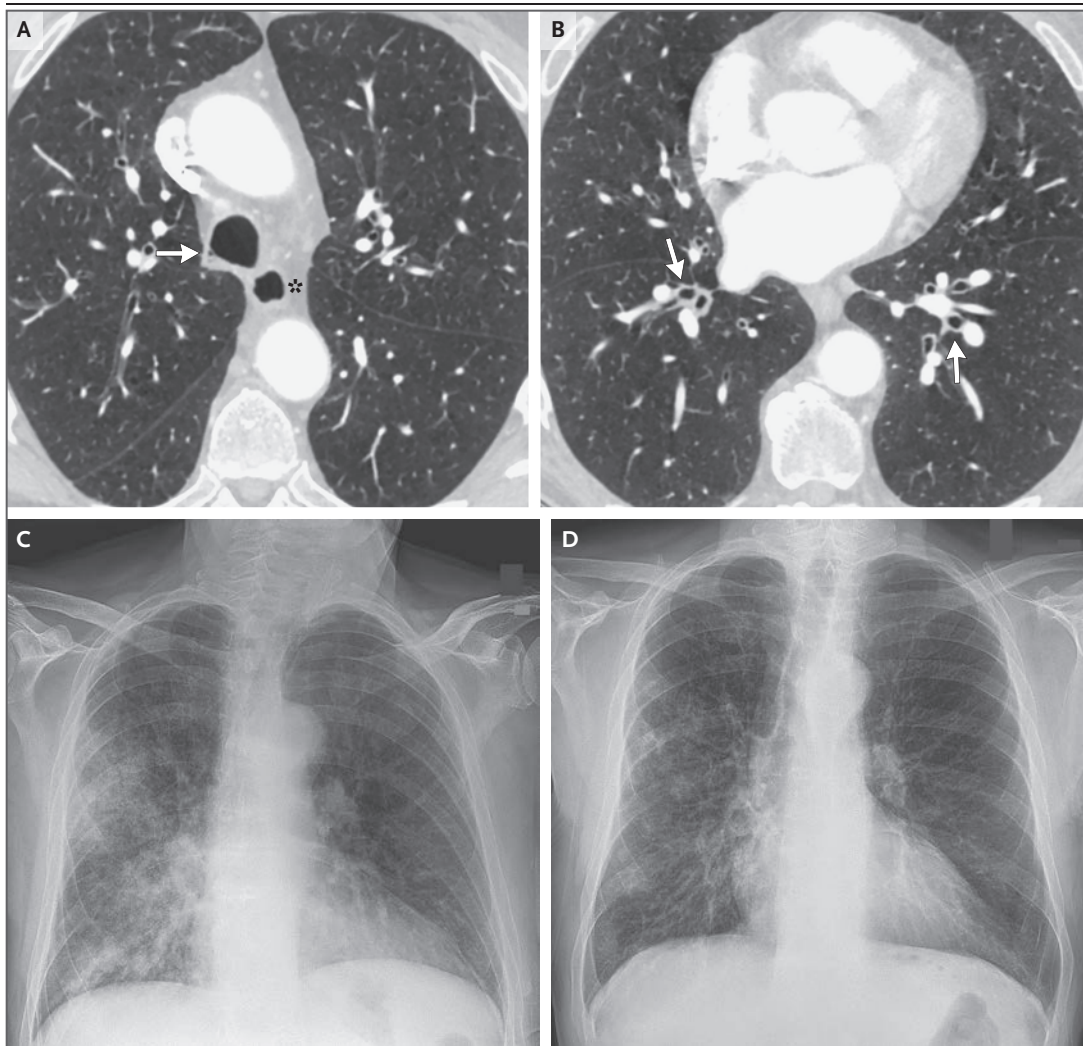


Figure 1. Initial Imaging Studies of the Chest.

An axial chest CT image obtained 5 months before the current admission (Panel A) shows layering secretions within the trachea (arrow), as well as a patulous esophagus (asterisk). An additional axial image from the same CT examination (Panel B) shows bronchial wall thickening (arrows). A frontal chest radiograph obtained 3 months before the current admission (Panel C) shows patchy and confluent opacities predominantly in the right mid and lower lung. A frontal chest radiograph obtained 1 month before the current admission (Panel D) shows patchy opacities in the right upper, middle, and lower lobes.

were normal, as were the results of liver-function and kidney-function tests; other laboratory test results are shown in Table 1.

Dr. Harvey Barnes: A chest radiograph (Fig. 1C) showed patchy and confluent opacities predominantly in the right mid and lower lung.

Dr. Alba: The patient was admitted to this hospital, and empirical treatment with intravenous vancomycin, cefepime, and levofloxacin was administered. A blood test for cryptococcal anti-

gen and a urine test for legionella antigen were negative. A sputum culture grew normal respiratory flora and *Candida albicans*; a test for *Pneumocystis jirovecii* was negative. During the next several days, fever resolved and the oxygen saturation and white-cell count normalized. On the fifth hospital day, the patient was discharged home to complete a 14-day course of levofloxacin. After he completed the course of levofloxacin, his cough and constitutional symptoms abated

Table 1. Laboratory Data.

Variable	Reference Range, Adults*	First Admission, 3 Mo before Current Admission	Second Admission, 1 Mo before Current Admission	Third Admission, 8 Days before Current Admission	Current Admission
Hemoglobin (g/dl)	13.5–17.5	15.4	14.7	12.4	13.6
Hematocrit (%)	41.0–53.0	44.5	46.2	40.9	42.8
White-cell count (per μ l)	4500–11,000	20,300	15,480	18,120	21,180
Differential count (%)					
Neutrophils	40–70	80.9	77.5	90.6	83.9
Lymphocytes	22–44	12.3	17.1	5.8	11.6
Monocytes	4–11	4.8	4.1	3.1	3.7
Eosinophils	0–8	0.4	0.8	0	0.3
Platelet count (per μ l)	150,000–400,000	255,000	199,000	480,000	406,000
Lactate (mmol/liter)†	0.5–2.2	2.8	—	2.6	1.8
Erythrocyte sedimentation rate (mm/hr)	0–13	16	—	54	—
C-reactive protein (mg/liter)	<8	126.6	—	149.7	—
1,3- β -D-glucan (pg/ml)	<60	488	—	420	>500
Venous blood gases					
Fraction of inspired oxygen	—	—	—	—	0.4
pH	7.30–7.40	—	—	—	7.36
Partial pressure of carbon dioxide (mm Hg)	38–50	—	—	—	64
Partial pressure of oxygen (mm Hg)	35–50	—	—	—	26

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

† To convert the values for lactate to milligrams per deciliter, divide by 0.1110.

partially but did not return to baseline. His primary care physician prescribed inhaled fluticasone–salmeterol and as-needed inhaled albuterol.

Six weeks after the patient was discharged from this hospital (approximately 5 weeks before the current admission), his wife began to have a nonproductive cough and rhinorrhea; several days later, the patient had similar symptoms, along with increased production of brown sputum. He sought medical attention after awakening with coughing and dyspnea. The temperature was 38.4°C and the oxygen saturation 89% while he was breathing ambient air.

Dr. Harvey Barnes: A chest radiograph (Fig. 1D) obtained 1 month before the current admission

showed patchy opacities in the right upper, middle, and lower lobes.

Dr. Alba: Electrolyte levels were normal, as were the results of liver-function and kidney-function tests; other laboratory test results are shown in Table 1. The patient was again admitted to this hospital, and empirical treatment with intravenous vancomycin, cefepime, and azithromycin; oral prednisone; and inhaled albuterol and ipratropium was administered. A sputum culture grew normal respiratory flora and *C. albicans*. After two days, the patient was discharged home to complete a 7-day course of levofloxacin.

Over the course of the next 3 weeks, the pa-

tient had malaise, productive cough, fever, and sweats. He stopped smoking cigarettes.

Eight days before the current admission, the patient was referred to the emergency department of this hospital for evaluation. On examination, the temperature was 37.2°C and the oxygen saturation 94% while he was breathing ambient air. There were crackles in both lung bases and in the right middle lung field, with no wheezes. Blood levels of electrolytes and lactate dehydrogenase were normal, as were the results of liver-function and kidney-function tests and urinalysis; other laboratory test results are shown in Table 1.

Dr. Harvey Barnes: A chest radiograph (Fig. 2A) showed multifocal reticulonodular and patchy opacities, which were confluent in the right lower lobe. CT of the chest (Fig. 2B and 2C), performed without the administration of intravenous contrast material, revealed multifocal clustered, centrilobular nodules and opacities in a tree-in-bud pattern with scattered areas of focal consolidation that were present predominantly in the right middle and lower lobes. There was associated mild mediastinal and right hilar lymphadenopathy that had not been present on the previous chest CT examination.

Dr. Alba: The patient was again admitted to this hospital, and additional testing was performed. Tests for antineutrophil cytoplasmic antibodies and anti-double-stranded DNA antibodies were negative, and blood levels of C3 and C4 were normal. A urine test for legionella antigen was negative.

On the third hospital day, a bronchoscopy and a transbronchial biopsy were performed. The bronchoalveolar-lavage (BAL) fluid obtained from the right middle lobe was cloudy; analysis of the fluid revealed 888 nucleated cells per microliter (of which 48% were neutrophils and 2% were lymphocytes, with no eosinophils). Gram's staining of the BAL fluid revealed neutrophils and sparse growth of gram-positive cocci. Staining for acid-fast organisms was negative; culture grew normal respiratory flora and *C. albicans*. Tests of the BAL fluid for influenza virus, adenovirus, parainfluenza virus, respiratory syncytial virus, human metapneumovirus, *P. jirovecii*, histoplasma antigen, and blastomyces antigen were negative, as were cultures for cytomegalovirus and herpes simplex virus. An interferon- γ release

assay for *M. tuberculosis* was also negative. The patient was discharged home on the third hospital day, with follow-up scheduled in the pulmonary clinic for 3 weeks later.

Four days after the patient was discharged, a BAL test for galactomannan returned a positive result (antigen index, ≥ 3.750 ; negative test, <0.5), and the patient was readmitted to this hospital. He reported ongoing dyspnea, cough with thick white sputum, and persistent fever with a temperature of up to 39.3°C. A review of systems was notable for diffuse chronic arthralgias, a bitter taste in the mouth, and intermittent dysphagia. There was no chest pain, edema, orthopnea, nausea, vomiting, jaundice, or rash, and he had no known current sick contacts. The weight was 5 kg lower than that recorded 1 year earlier.

Medical history was notable for polyarteritis nodosa, deep venous thrombosis, gastroesophageal reflux disease (GERD), pancreatitis in the context of choledocholithiasis, hypertension, dyslipidemia, diabetes mellitus, osteopenia, and chronic neuropathic pain. Medications included aspirin, prednisone, methotrexate, trimethoprim-sulfamethoxazole, folate supplementation, metformin, atenolol, simvastatin, pregabalin, and as-needed oxycodone, as well as inhaled fluticasone-salmeterol and albuterol as needed. Lisinopril had reportedly caused pancreatic inflammation; there were no other known adverse reactions to medications. The patient was of White European ancestry. He lived in New England with his wife and was retired from an office position. Three decades earlier, he had worked in a machine shop where solvents, fuels, and sulfur compounds were used but had had no known exposure to these agents. There was no history of exposure to mold, pets, or asbestos. He had received influenza and pneumococcal vaccines. He and his wife had traveled within the United States and to the Caribbean. He had smoked 1.5 packs of cigarettes per day for 30 years but had quit in the weeks before the current admission. He had previously smoked marijuana, rarely consumed alcohol, and never used other substances. There was no family history of autoimmune or pulmonary disease; his father had died from colorectal cancer, and his mother had had esophageal cancer and had died after a stroke. His three adult children were healthy.

The temperature was 36.6°C, the heart rate

75 beats per minute, the blood pressure 135/69 mm Hg, the respiratory rate 28 breaths per minute, and the oxygen saturation 91% while the patient was receiving supplemental oxygen through a nasal cannula at a rate of 6 liters per minute. He had a slightly cushingoid appearance and was tachypneic, although he was able to speak in full sentences. Auscultation of the chest revealed rales at the right lung base, rhonchi during both the inspiratory and expiratory phases that were more pronounced in the right lung than in the left lung, and intermittent inspiratory wheezing in the right lung. The rest of the examination was normal.

Dr. Harvey Barnes: A chest radiograph (Fig. 2D) showed progression of the diffuse bilateral reticulonodular opacities, with additional patchy opacities in the right lung and left lower lobe, as compared with the image obtained 1 week earlier.

Dr. Alba: Blood levels of electrolytes, amylase, lipase, N-terminal pro-B-type natriuretic peptide, and troponin T were normal, as were the results of tests of kidney and liver function; other laboratory test results are shown in Table 1. Urinalysis was normal, and a urine test for legionella antigen was negative.

The BAL-fluid culture obtained during the previous admission grew enterococcus species, *C. albicans*, and *C. glabrata*; no mycobacteria grew in any of three induced sputum samples obtained during the previous admission. Empirical treatment with intravenous voriconazole was initiated, and ongoing treatment with methotrexate was interrupted.

Cytologic analysis of the BAL fluid and the transbronchial-biopsy specimen obtained during the previous admission revealed evidence of acute inflammation, bronchial columnar cells, and pulmonary macrophages. There were no malignant cells. Evaluation of the bronchial mucosa revealed no abnormalities.

On the second hospital day, the temperature was 37.9°C and the oxygen saturation 91% while the patient was receiving supplemental oxygen through a nasal cannula at a rate of 5 liters per minute.

Dr. Harvey Barnes: Repeat CT of the chest (Fig. 2E and 2F) was performed after the administration of intravenous contrast material as part of a pulmonary embolism protocol. This imaging study revealed extensive centrilobular pul-

monary nodules, opacities in a tree-in-bud pattern, and patchy ground-glass opacities. There was bronchial wall thickening with mucus plugging, predominantly affecting the lower lobes. There were additional areas of nodular consolidation that were more prominent in the right lung than in the left lung. These findings indicated a progression of disease as compared with the chest CT performed 1 week before the current admission. There was no evidence of pulmonary embolism.

Dr. Alba: Additional induced sputum samples were obtained for culture; Gram's staining revealed abundant neutrophils and abundant mixed gram-positive and gram-negative organisms.

On the third hospital day, the temperature was 36.9°C and the oxygen saturation 96% while the patient was receiving supplemental oxygen through a nasal cannula at a rate of 3 liters per minute.

Diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Rachel S. Knipe: This 57-year-old immunosuppressed man with polyarteritis nodosa presents with a waxing and waning pneumonia syndrome that is characterized by centrilobular nodules and tree-in-bud nodularity on chest imaging. Polyarteritis nodosa is a medium-vessel vasculitis that can affect the kidneys, skin, nerves, muscles, and gastrointestinal tract but notably does not cause disease in the lungs. Therefore, I will develop a differential diagnosis based on the high likelihood that this patient's condition is caused by an alternative process, and I will rely heavily on the radiographic findings to narrow the list of possible diagnoses.

INFECTION

This patient's presentation is not consistent with a typical bacterial pneumonia, but pneumonia caused by an atypical organism such as mycoplasma should be considered. Although mycoplasma pneumonia is typically characterized by patchy reticulonodular and ground-glass opacities, it can also be manifested by centrilobular nodules and tree-in-bud nodularity. However, this patient's symptoms did not resolve after he received levofloxacin, which is highly effective in treating mycoplasma; thus,

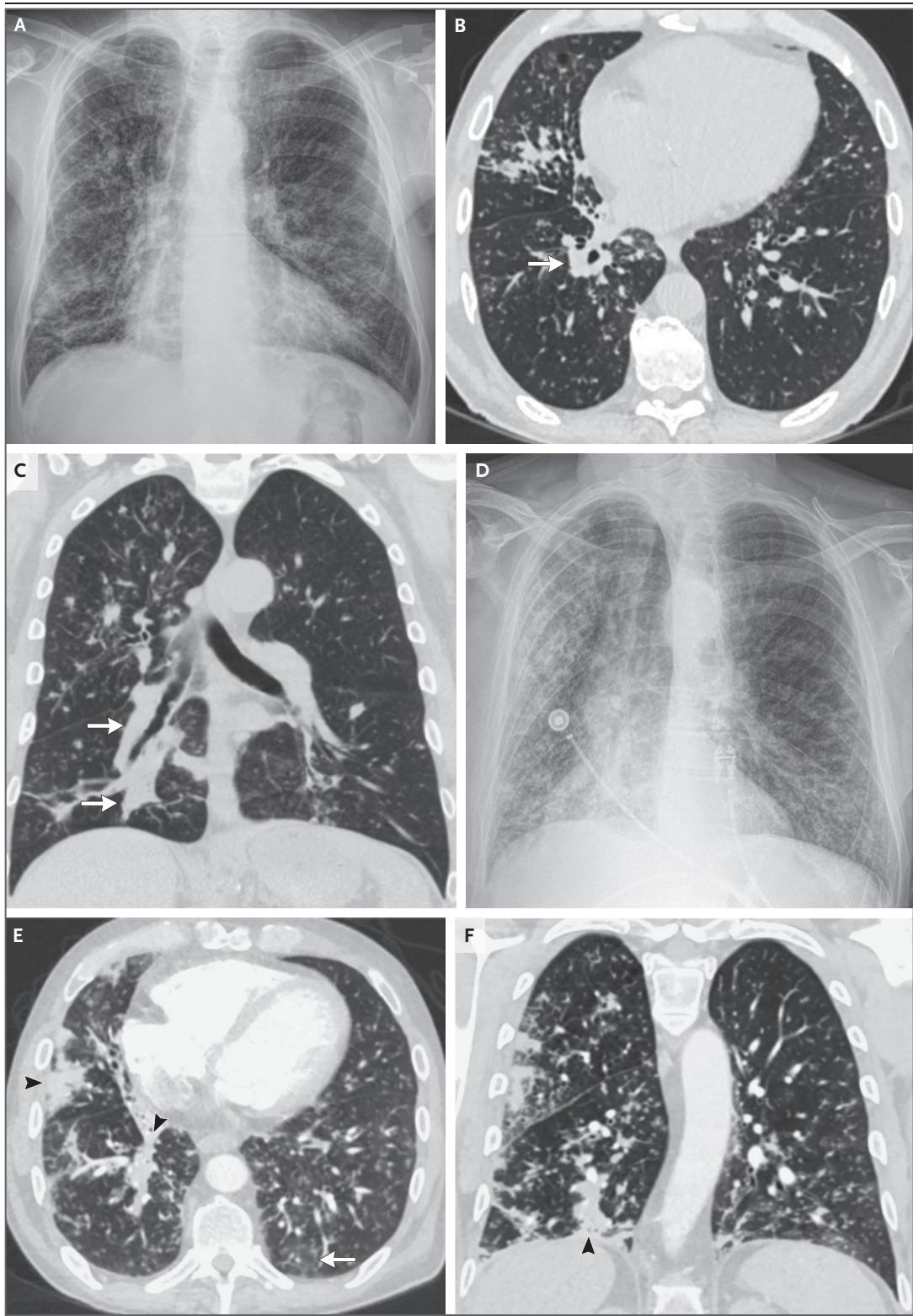


Figure 2 (facing page). Subsequent Imaging Studies of the Chest.

A frontal chest radiograph obtained 1 week before the current admission (Panel A) shows multifocal reticulonodular and patchy opacities, which are confluent in the right lower lobe. Axial (Panel B) and coronal (Panel C) chest CT images obtained 1 week before the current admission show new multifocal clustered, centrilobular nodules and tree-in-bud nodularity; there are scattered areas of consolidation (arrows) that are most prominent in the right middle and lower lobes. A frontal chest radiograph obtained during the current admission (Panel D) shows progression of the diffuse reticulonodular opacities bilaterally, with additional patchy opacities throughout the right lung and left lower lobe, as compared with the previous chest radiograph obtained 1 week earlier. Axial (Panel E) and coronal (Panel F) chest CT images obtained during the current admission show increased diffuse centrilobular nodules and tree-in-bud nodularity, patchy ground-glass opacities (arrow), and additional areas of consolidation (arrowheads) that are more prominent in the right lung than in the left lung; these findings indicate a progression of disease, as compared with the chest CT images obtained 1 week before the current admission (Panels B and C).

mycoplasma pneumonia is an unlikely diagnosis.

Nocardia is a slow-growing modified acid-fast bacterium with tropism for the lungs and brain. Risk factors include immunosuppression, human immunodeficiency virus (HIV) infection, cancer, and diabetes. Radiographic imaging studies typically show single or multiple pulmonary nodules in the upper lobes that may cavitate. However, this patient had multiple sputum and BAL-fluid cultures that were negative, which makes a diagnosis of nocardiosis unlikely.

Fungal infection, particularly due to aspergillus, should be considered because of the positive BAL test for galactomannan. The galactomannan test recognizes a polysaccharide component of the fungal cell wall. The test is only 81% specific for invasive aspergillus, with positive tests often occurring in the presence of other fungi that share cross-reacting antigens such as fusarium or penicillium.¹ False positive results can also occur because of fungal colonization in the airways, aspiration of contaminated food particles, contaminated BAL fluid or blood products, or the use of antibiotic agents such as piperacillin-tazobactam.² Although ubiquitous

in nature, aspergillus species tend to cause invasive disease in the presence of severe immunosuppression. The lung is the most common site of infection, and chest imaging often shows multiple cavitory nodules. Aspergillus is angioinvasive, and nodules are often surrounded by a “halo” of ground glass, a finding that indicates hemorrhage into surrounding tissue. Despite the positive test for galactomannan, the radiographic findings in this case are not suggestive of aspergillus.

P. jirovecii pneumonia is an opportunistic fungal infection that is most commonly seen in immunocompromised patients. This patient received prophylactic trimethoprim-sulfamethoxazole, which markedly reduces the risk of *P. jirovecii* pneumonia and nocardia infection. Typical radiographic findings include diffuse bilateral interstitial infiltrates with ground-glass opacities, which often cavitate and become cystic. Although this patient’s blood 1,3- β -D-glucan test was positive, a finding that is often suggestive of *P. jirovecii* pneumonia, his imaging findings are not typical of this condition, and the BAL fluid and several induced sputum specimens were negative for *P. jirovecii*; therefore, this diagnosis can be ruled out.

Candida is part of the normal flora of the gastrointestinal and genitourinary tracts, but invasive disease occurs in the context of severe immunosuppression. Candida pneumonia is rarely diagnosed; because the majority of candida growth in BAL fluid is correctly attributed to colonization, it is difficult to identify cases of pathogenic growth of candida in the lung without histologic confirmation.³ Lung biopsy could identify invasive disease, but it is rarely performed in such cases. Imaging typically shows bilateral nodules and consolidation, but centrilobular nodules and tree-in-bud nodularity have been reported. In this case, there is evidence of candida colonization identified from specimens obtained during bronchoscopy and sputum sampling, and the blood 1,3- β -D-glucan test was positive; therefore, invasive candida pneumonia cannot be ruled out.

The 1,3- β -D-glucan test is less specific for aspergillus than the galactomannan test, because this compound is present in the cell wall

of many yeasts and molds, such as *P. jirovecii*, candida species, fusarium, and acremonium. False positive test results can also occur in the presence of cellulose membrane, which is used for hemodialysis, intravenous immune globulin treatment, and albumin therapy and is also found in surgical gauze.

The radiographic abnormalities seen in this patient are also consistent with mycobacterial disease. The lungs are a primary site of infection in patients with tuberculosis; however, this patient had no known exposure to a person with active tuberculosis, and multiple mycobacterial cultures and tests for latent tuberculosis were negative. Infection with nontuberculous mycobacterium, which often complicates structural lung diseases such as bronchiectasis, should be considered; however, all this patient's mycobacterial cultures were negative. Viruses such as respiratory syncytial virus, influenza virus, parainfluenza virus, and adenovirus can cause a diffuse centrilobular and tree-in-bud nodular pattern on imaging. However, viral studies of the BAL fluid were negative.

INTERSTITIAL LUNG DISEASE

Given that no fibrotic changes were noted on chest imaging, I narrowed my focus to inflammatory interstitial lung diseases (ILDs) that can cause centrilobular nodules. Hypersensitivity pneumonitis, an immune response in the lungs to inhaled antigens, can have a waxing and waning clinical course, since symptoms flare with antigen exposure. However, the imaging findings in this case are only partially consistent with a diagnosis of hypersensitivity pneumonitis. Centrilobular nodules are classically associated with hypersensitivity pneumonitis, but tree-in-bud nodularity is not. In this patient, there was no evidence of mosaic attenuation or air trapping and no clear environmental exposure; there was no lymphocytosis identified on analysis of the BAL fluid, which is characteristic of — although not specific for — hypersensitivity pneumonitis.

Similar to hypersensitivity pneumonitis, sarcoidosis is a granulomatous reaction and can involve any part of the body, with the lungs, skin, joints, and eyes being most affected. However, the imaging findings in this case are not consistent with a diagnosis of sarcoidosis, because the nodules are not in a perilymphatic

distribution and there is no clinically significant hilar lymphadenopathy.

Organizing pneumonia is an inflammatory reaction in the lung to injury that can be due to infection, connective-tissue disease, cancer, or medications, or it can be cryptogenic. This patient's waxing and waning clinical course is consistent with organizing pneumonia, but imaging typically shows peripheral and peribronchovascular nodular consolidations. Respiratory bronchiolitis-associated ILD is a smoking-related ILD that is manifested by dyspnea but not by fever and productive cough, which were seen in this patient. This patient's imaging is also not fully consistent with a diagnosis of respiratory bronchiolitis-associated ILD, since patients usually present with a predominant pattern of air trapping and ground-glass opacities and, less commonly, centrilobular nodules.

Several lymphoproliferative ILDs, including lymphocytic interstitial pneumonia, should also be considered. Lymphocytic interstitial pneumonia is often associated with Sjögren's syndrome or HIV type 1 (HIV-1) infection and is characterized by the presence of cysts on imaging. Follicular bronchiolitis is a lymphoid proliferation of bronchus-associated lymphoid tissue that is also usually associated with Sjögren's syndrome or HIV-1 infection. Although this patient's imaging findings, which included centrilobular nodules and tree-in-bud nodularity, are consistent with follicular bronchiolitis, his clinical presentation, which included fever and productive cough, is not. Diffuse panbronchiolitis is rarely diagnosed in persons who are not of Asian ancestry, and this patient was of White European ancestry.

DRUG-RELATED TOXIC EFFECT

This patient received treatment with cyclophosphamide and methotrexate. Reports of pulmonary toxic effects associated with the use of cyclophosphamide are rare; however, when acute pneumonitis occurs, it usually develops within the first 6 months of treatment, and when chronic pneumonitis occurs, it is characterized by fibrosis, which was not seen on this patient's imaging studies. Methotrexate can cause a diffuse pneumonitis, but this usually occurs during the first year of treatment. None of these features fit the temporal or radiographic pattern described in this case.

CANCER

It is important to consider cancer in any case involving persistent lung opacities. Primary lung cancer is typically manifested by discrete nodules and lymphadenopathy. Lymphomatoid granulomatosis, a rare pulmonary lymphoma that is associated with Epstein–Barr virus, has been reported to occur in the context of methotrexate therapy. Imaging typically shows single or multiple nodules that can cavitate in a peribronchovascular distribution. Tumor embolization is characterized by nodules in a random distribution, and lymphangitic carcinomatosis is characterized by nodules in a perilymphatic distribution. None of these diagnoses fit with this patient's radiographic pattern.

RECURRENT ASPIRATION

In pulmonary medicine, we have a saying that “everyone aspirates.” Aspiration can be clinically silent and can occur even in the absence of overt reflux symptoms. The centrilobular nodules and tree-in-bud pattern of opacities seen on this patient's imaging studies are consistent with aspiration. In addition, this patient had secretions in the airways, mucus plugging, a hiatal hernia, and some residual contrast material in the esophagus, all of which are suggestive of aspiration of oral and gastric contents into the airways. His clinical history, especially the history of GERD and a bitter taste in the mouth, suggests the occurrence of aspiration. This patient's waxing and waning clinical course would also be consistent with aspiration into the airways. The BAL fluid was notable for acute inflammation and the presence of squamous cells, candida, and bacteria, findings that are consistent with — although not specific for — aspiration of oral contents. Given this presentation, additional testing is warranted.

On the basis of this patient's clinical presentation and radiographic evaluation, infectious and aspiration bronchiolitis are the most likely diagnoses in this case. A thorough microbiologic workup was negative except for the fungal markers. The clinical and radiographic presentations are most consistent with a diffuse aspiration bronchiolitis; however, given this patient's degree of immunosuppression and positive fungal markers, I would proceed with a lung biopsy to rule out fungal pneumonia and to confirm the diagnosis of aspiration bronchiolitis.

DR. RACHEL S. KNIPE'S DIAGNOSIS

Diffuse aspiration bronchiolitis.

PATHOLOGICAL DISCUSSION

Dr. Lida P. Hariri: The patient underwent video-assisted thoracoscopic surgical wedge biopsy of the right upper and middle lobes. Histologic examination of the biopsy specimen (Fig. 3) revealed evidence of multifocal organizing pneumonia with prominent granulomatous inflammation associated with giant cells. Numerous foci of peribronchiolar histiocytes and giant cells formed intraluminal polypoid structures that obstructed the bronchiolar lumina. Rare foci of foreign material that were consistent with aspirated food particles were identified. There was associated focal microabscess formation and organizing fibrosis. Fibrinous pleuritis was also present. The background lung parenchyma showed numerous hemosiderin-laden macrophages within the alveolar spaces and associated interstitial fibrosis. There was no evidence of vasculitis. Gomori methenamine–silver, Brown–Hopps, and acid-fast stains for microorganisms were negative. This combination of histologic features is consistent with the diagnosis of aspiration pneumonia.

PATHOLOGICAL DIAGNOSIS

Aspiration pneumonia.

ADDITIONAL IMAGING STUDIES

Dr. Harvey Barnes: The patient underwent a video-fluoroscopic swallowing examination in collaboration with a speech–language pathologist. The examination revealed moderate pharyngeal dysphagia with delayed initiation of swallowing. There was deep laryngeal penetration of nectar and thin liquid consistencies with occasional trace silent aspiration of these consistencies before and during swallowing.

The patient subsequently underwent a barium-swallow study, which showed normal esophageal motility. When the patient was in the prone position, a small amount of contrast material was aspirated into the trachea and right bronchial tree (Fig. 4A). A small sliding hiatal hernia was identified (Fig. 4B) when the patient was in

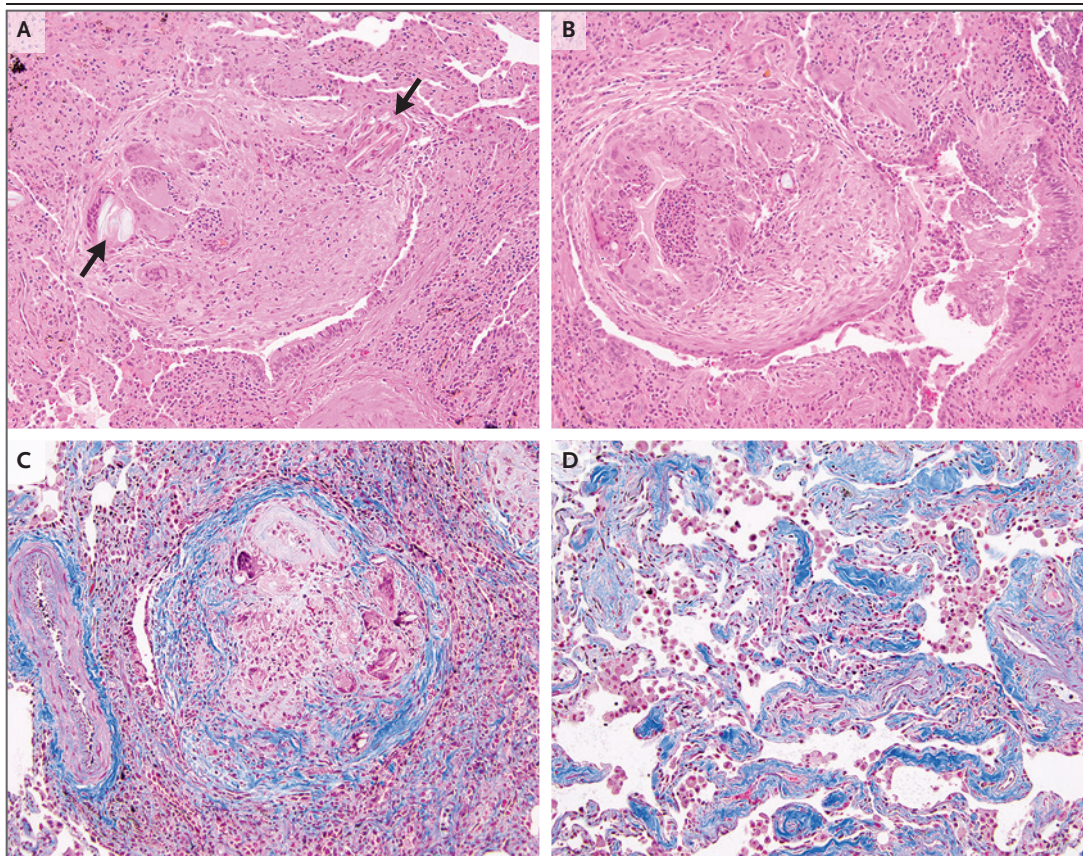


Figure 3. Biopsy Specimens of the Lung.

Hematoxylin and eosin staining (Panels A and B) and Masson trichrome staining (Panel C) of specimens obtained from the surgical wedge biopsy show evidence of multifocal organizing pneumonia with prominent granulomatous inflammation. Numerous foci of peribronchiolar histiocytes and giant cells form intraluminal polypoid structures that obstruct the bronchiolar lumina. Rare foreign material (Panel A, arrows) consistent with aspirated food particles is present. There is associated focal microabscess formation (Panel B) and organizing fibrosis (Panel C). Masson trichrome staining (Panel D) shows that the background lung parenchyma contain numerous hemosiderin-laden macrophages within the alveolar spaces with associated interstitial fibrosis. There is no evidence of vasculitis. This combination of histologic features is consistent with aspiration pneumonia.

the prone and supine positions. No hiatal hernia was seen when the patient was in the upright position. Moderate gastroesophageal reflux up to the level of the carina was seen when the patient was in the prone and supine positions.

DISCUSSION OF MANAGEMENT

Dr. Alba: Aspiration of oropharyngeal or gastric contents into the airways is a common yet underappreciated cause of pulmonary disease.⁴ Half of normal adults have been found to aspirate oropharyngeal contents into their lungs while asleep.⁵ Risk factors for aspiration include a de-

pressed level of consciousness, an abnormal swallowing mechanism, increased gastroesophageal reflux, and an impaired cough reflex.⁶ This patient had multiple risk factors for aspiration pneumonia, including GERD with a hiatal hernia; opioid use, which can impair the level of consciousness and gastroesophageal motility and can suppress cough; and extensive tobacco use with undiagnosed chronic obstructive pulmonary disease. Diffuse aspiration bronchiolitis, the condition that was ultimately diagnosed in this patient, is a disease characterized by the often-occult aspiration of gastric content into the airways.⁷

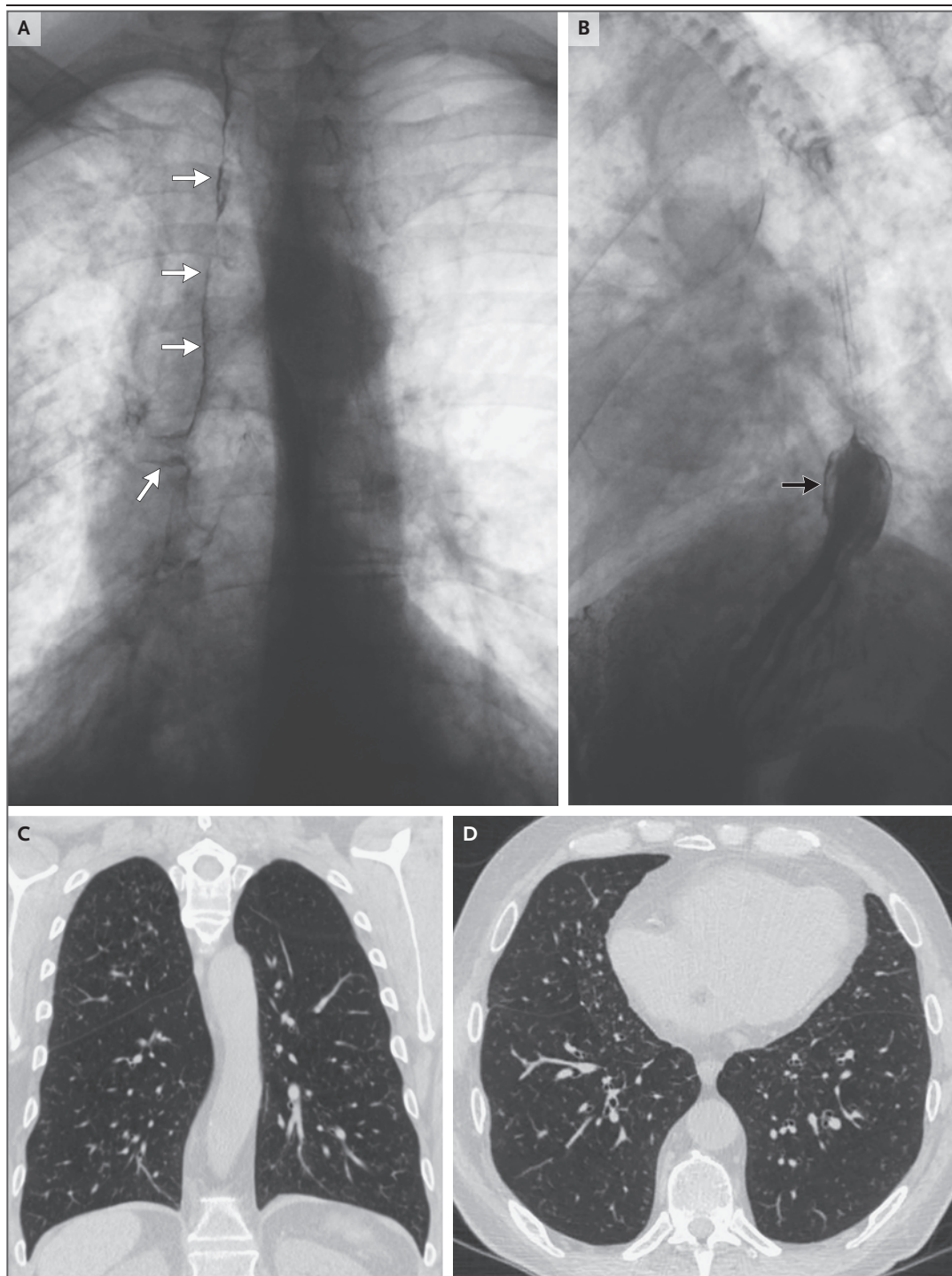


Figure 4. Barium-Swallow Study and Additional Imaging Studies of the Chest.

Fluoroscopic spot images of a barium-swallow examination show evidence of aspiration of contrast material, which outlines the tracheobronchial tree (Panel A, arrows), and a small sliding hiatal hernia (Panel B, arrow). Coronal (Panel C) and axial (Panel D) CT images of the chest obtained 6 months after the current admission show a decrease in nodular and tree-in-bud opacities, as well as resolution of ground-glass opacities and consolidations.

After the diagnosis of aspiration was established, the patient completed a course of antimicrobial therapy for aspiration pneumonia, began high-dose pharmacologic therapy to suppress gastric acidity, and was evaluated by the pain service to transition to a nonopioid pain regimen. He was also counseled regarding dietary and lifestyle modifications to reduce the occurrence of gastroesophageal reflux; these modifications included sleeping with the head of the bed elevated with the use of a mattress wedge.

With the implementation of aggressive lifestyle modifications and acid blockade, the patient had near-complete resolution of his pulmonary symptoms, no longer required supplemental oxygen therapy, and had normalization of his

inflammatory markers. Six months after the current admission, repeat pulmonary-function testing revealed clinically significant improvements. In addition, follow-up CT of the chest (Fig. 4C and 4D) revealed substantial decreases in the number of nodules and in the number of opacities with a tree-in-bud pattern, as well as resolution of consolidations and patchy ground-glass opacities.

FINAL DIAGNOSIS

Aspiration pneumonia.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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